

Data Structures Lab (CL2001)

Date: December 15th 2025

Instructor(s)

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Final Exam

Total Time (Hrs): 2 hrs
Total Marks: 40
Total Questions: 3

Roll No

Section

Student Signature

Do not write below this line

Clearly mention the assumptions if made.

STL is allowed.

Attempt all the questions.

CLO #: 1

[10 Marks] Q1: You are given a queue of integers implemented as `std::queue<int>` in C++. Your task is to reverse all elements of the queue using recursion. To do so, implement the following function:

`void reverseQueue(std::queue<int>& q);`

- The function should reverse the entire queue in-place.
- You may only use recursion and standard queue operations:
 - `q.push()`
 - `q.pop()`
 - `q.front()`
 - `q.empty()`

No loops or extra data structures are allowed.

Sample Run:

Input: 10 20 30 40 50

Output: 50 40 30 20 10

CLO #: 2

[10 Marks] Q2: Given an array of characters, the goal is to rearrange the elements so that no two identical characters appear consecutively. Before attempting the rearrangement, you must first determine whether it is feasible.

Feasibility condition: The rearrangement is impossible if the maximum frequency of any character exceeds $(n + 1) / 2$, where n is the total number of elements in the array.

Approach to generate a valid arrangement (if possible):

1. Use a max heap (priority queue) to always select the character with the highest remaining frequency.

2. Keep track of the previously placed character and ensure it is not placed consecutively.
3. Repeat this process until all characters are placed.

Example:

For the input {'a', 'a', 'a', 'b', 'b', 'c', 'c'}, one valid rearrangement is abacabc

CLO #: 2

[15 Marks] Q3: You are given a queue of integers, each representing the priority of a task. Your goal is to reorder the queue in a zig-zag pattern:

- The first element should be the smallest integer.
- The second element should be the largest integer.
- The third element should be the second smallest,
- The fourth element should be the second largest, and so on...

Essentially, the reordered queue alternates between next smallest and next largest elements.

Constraints:

- You may use two stacks (`std::stack<int>`) in addition to the original queue.
- No additional arrays, vectors, or other containers are allowed.
- Must work for any queue size ≥ 1 (even or odd).

Prototype of the function:

```
void zigzagQueue(std::queue<int>& q);
```

Example:

Input: [4, 1, 6, 3, 5, 2]

Output: [1, 6, 2, 5, 3, 4]

[5 Marks] Write a menu-driven main program that allows the user to select and test each function on the desired input interactively.

Menu:

1. Reverse a queue using recursion
2. Rearrange characters to avoid consecutive duplicates
3. Zig-zag reorder a queue
4. Exit

Enter your choice:

Best of Luck!!!